

MUHAMMAD HANZALA KHAN

AI / ML Engineer — Computer Vision · NLP · Generative AI · Agentic AI

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PROFESSIONAL SUMMARY

BS Artificial Intelligence student at Pak-Austria Fachhochschule Institute of Applied Sciences & Technology (PAF-IAST) with five internships spanning computer vision, NLP, agentic AI systems, and applied machine learning. Built and deployed systems including an agentic multi-disease chest X-ray screening pipeline (LangGraph, RAG, Qdrant, Grad-CAM), a TensorFlow/MobileNetV2 factory safety-monitoring pipeline with a Streamlit dashboard, and a PyTorch-based leukemia subtype classifier with a RAG explanation engine orchestrated through n8n, alongside multiple emotion-aware and NLP applications using LangChain, Flask, and OpenCV. Seeking an AI/ML engineering internship or research role applying practical experience in Python, deep learning, agentic and RAG-based systems, and full-stack AI deployment.

EDUCATION

BS Artificial Intelligence, Pak-Austria Fachhochschule Institute of Applied Sciences & Technology (PAF-IAST), Haripur **2023 – Present**

Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, NLP, Data Structures, Databases, Computer Networks

ICS, Base Group of College, Wah Cantt

2021 – 2023

Matriculation, Sir Syed High School & College, Wah Cantt

2019 – 2021

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, HTML/CSS

AI / ML: Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI, Agentic AI Systems, RAG (Retrieval-Augmented Generation), ANN/RNN/LSTM, CNNs (Transfer Learning), Affective (Emotional) AI, Federated Learning, Explainable AI (XAI / Grad-CAM)

Frameworks & Libraries: TensorFlow, Keras, PyTorch, OpenCV, Scikit-learn, LangChain, LangGraph, Flask, FastAPI, Streamlit

AI Infrastructure: Vector Databases (ChromaDB, Qdrant), LLM APIs (Groq), Local LLMs (Ollama), n8n Workflow Automation, Docker

Databases & Tools: SQLite, MySQL, Git, GitHub, VS Code, Jupyter Notebook, Google Colab

PROFESSIONAL EXPERIENCE

AI Engineering Intern

Aug – Sep 2026

Mediatiz Foundation, On-site

- Built an Agentic Radiology Assistant — a multi-disease chest X-ray screening pipeline combining two DenseNet121 vision models, Grad-CAM explainability, and a RAG-grounded medical knowledge base.
- Designed a single LangGraph agent workflow orchestrating vision inference, retrieval (Qdrant), and report generation via a locally hosted LLM (Ollama).
- Expanded the system from single-condition to six-condition screening per stakeholder feedback, using a "specialist + generalist" dual-model architecture to preserve reliability on the primary condition.

AI & ML Intern

Jun – Aug 2025

ITSOLERA Pvt Ltd, Islamabad

- Built a real-time facial-emotion detection system that classifies user mood from live video, powering a mood-based music recommendation feature.
- Developed a Factory Compliance Monitoring System, combining computer vision (TensorFlow, MobileNetV2) with policy rule parsing (PyMuPDF) to detect workplace safety violations and report them through an interactive Streamlit dashboard.
- Contributed to a Brand Sentiment Monitor and Scholarship Finder platform, applying NLP, Flask, and MySQL, alongside ARIMA-based time-series forecasting for Ethereum price prediction.

AI & ML Intern

Jun – Jul 2025

Arch Technologies, Islamabad

- Built deep learning models for medical image analysis, including brain tumor segmentation and breast cancer classification.
- Strengthened model-building and deployment workflows through applied AI/ML project development.

Web Programmer Intern

Jul – Aug 2024

Pakistan Ordnance Factories, Wah Cantt

- Developed web applications and automation scripts using Python and C++.

PROJECTS

Agentic Radiology Assistant — Multi-Disease X-ray Screening (LangGraph + RAG + Grad-CAM)

- Trained a Pneumonia-specialist DenseNet121 (0.92 ROC-AUC) and a second multi-label DenseNet121 covering 5 additional chest conditions (NIH ChestX-ray14), using class-weighted loss and per-disease threshold tuning.
- Built a Grad-CAM explainability layer that reports both the location and visual pattern behind each finding, in plain language, for non-expert readers.
- Designed a RAG pipeline (Qdrant + sentence-transformers) over a 12-document curated medical knowledge base, with every report claim traceable to a source.
- Orchestrated the full pipeline — dual vision models, retrieval, and report generation — as a single LangGraph agent, served through a Streamlit interface with a local LLM (Ollama).

Leukemia Subtype Classifier — CNN + RAG + n8n Automation Pipeline

- Trained a ResNet18 CNN (PyTorch, transfer learning) to classify Acute Lymphoblastic Leukemia subtypes from blood smear images, achieving 99.6% test accuracy; used Grad-CAM for explainability.
- Built a RAG explanation engine (ChromaDB vector store + Groq LLM API) that generates natural-language, context-grounded explanations for each prediction.
- Deployed the model and RAG pipeline as a FastAPI service, then orchestrated the full flow end-to-end with a self-hosted n8n (Docker) workflow — image upload, classification, and explanation delivered through a single automated pipeline.

File Management System

- Built a GUI-based file management tool in Python supporting 18+ operations (create, delete, rename, search, move), evolved from an initial C and Bash script version.

CERTIFICATIONS

- AI For Everyone — DeepLearning.AI
- Supervised Machine Learning: Regression and Classification — Stanford Online / DeepLearning.AI

LANGUAGES

Urdu (Native) | English (Proficient)